CHAPTER D

Curriculum coverage matrix

The expectations are rephrased and broken down into verifiable entries. “Covered” indicates a localised resource, not a mastery attested for the student. Not all of the optional extension items in the BO (Bulletin officiel) are compulsory. The Year 11 (Seconde) resources are maintained in R001–R030 and STATS. The complete CSV table provides the four final-pagination columns and the lesson, example and exercise identifiers.

ID	Expectation / type	Lesson, example	Exercise
BO001	Membership and inclusion (Content)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO002	Empty set, union, intersection and complement (Content)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO003	Sets of numbers and intervals (Content)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO004	Ordered pair and Cartesian product (Content)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO005	Cardinality of a finite set (Content)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO006	Connectives and/or (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO007	Counterexample (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO008	Implication and equivalence (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO009	Converse and contrapositive (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO010	Necessary and sufficient condition (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO011	Identity, equation, variable, unknown and parameter (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO012	Quantifiers and implicit quantification (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO013	Negation of quantified statements (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO014	Proof by cases (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO015	Proof by contradiction (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO016	Proof by contrapositive (Skill)	LOGIQUE, p. 116 ; LOGIQUE	LOG-EX, p. 116
BO017	Variables and assignment (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO018	Functions and return values (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO019	Bounded loops and unbounded loops with a stopping condition (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO020	Conditional statements (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO021	Lists defined in extension, successive appends and list comprehension (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO022	Manipulating, adding and removing elements and indices (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118

ID	Expectation / type	Lesson, example	Exercise
BO023	Traversing a list and iterating over its elements (Skill)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO024	Direct rate (Skill)	R006, p. 120 ; R006	R006, p. 120
BO025	Initial value (Skill)	R007, p. 121 ; R007	R007, p. 121
BO026	Percentage rate of change (Skill)	R008, p. 121 ; R008	R008, p. 121
BO027	Successive changes (Skill)	R009, p. 121 ; R009	R009, p. 121
BO028	Reverse change (Skill)	R010, p. 121 ; R010	R010, p. 121
BO029	Zero-product property (Skill)	R011, p. 121 ; R011	R011, p. 121
BO030	Sign of an affine expression (Skill)	R014, p. 122 ; R014	R014, p. 122
BO031	Sign of a factorised quadratic expression (Skill)	C02-U4, p. 15 ; C02-EX4	C02-E1, p. 16
BO032	Expanding, factorising and simplifying (Skill)	R012, p. 121 ; R012	R012, p. 121
BO033	Graphical equations and inequalities (Skill)	R078, p. 132 ; R078	R078, p. 132
BO034	Graphical sign and variations (Skill)	R078, p. 132 ; R078	R078, p. 132
BO035	Drawing a line from an equation or from a point and a gradient (Skill)	R016, p. 122 ; R016	R016, p. 122
BO036	Reading the equation of a line (Skill)	R015, p. 122 ; R015	R015, p. 122
BO037	Gradient from two points (Skill)	R015, p. 122 ; R015	R015, p. 122
BO038	Charts, histograms, bar charts, pie charts and box plots (Skill)	STATS, p. 118 ; STATS	STATS-EX, p. 118
BO039	Moving from data to a representation (Skill)	STATS, p. 118 ; STATS	STATS-EX, p. 118
BO040	Statistical measures (Skill)	STATS, p. 118 ; STATS	STATS-EX, p. 118
BO041	Conditional probabilities in a table or tree (Skill)	C05-U1, p. 47 ; C05-EX1	C05-E1, p. 49
BO042	Distinguishing intersection from two conditionings (Skill)	C05-U1, p. 47 ; C05-EX1	C05-E1, p. 49
BO043	Explicit, recursive, algorithmic modes and patterns (Content)	C01-U1, p. 2 ; C01-EX1	C01-E1, p. 4
BO044	Notation for the sequence and for a term (Content)	C01-U1, p. 2 ; C01-EX1	C01-E1, p. 4
BO045	Moving between natural language, algebra and graph (Skill)	C01-U1, p. 2 ; C01-EX1	C01-E5, p. 5
BO046	Modelling a pattern or a count (Skill)	C01-U1, p. 2 ; C01-EX1	C01-E5, p. 5
BO047	Calculating explicit, recursive or algorithmic terms (Skill)	C01-U1, p. 2 ; C01-EX1	C01-E5, p. 5
BO048	Arithmetic sequence, general term, constant increase, affine link (Content)	C01-U2, p. 2 ; C01-EX2	C01-E2, p. 5
BO049	Geometric sequence, general term, constant rate, exponential link (Content)	C01-U3, p. 3 ; C01-EX3	C01-E3, p. 5
BO050	General term of an arithmetic sequence (Proof)	C01-U2, p. 2 ; C01-EX2	C01-E8, p. 6
BO051	General term of a geometric sequence (Proof)	C01-U3, p. 3 ; C01-EX3	C01-E3, p. 5
BO052	Sum of the first integers (Proof)	C01-U4, p. 3 ; C01-EX4	C01-E4, p. 5
BO053	Sum of 1 to q to the power n (Proof)	C01-U4, p. 3 ; C01-EX4	C01-E4, p. 5
BO054	Calculating sums of consecutive terms (Skill)	C01-U4, p. 3 ; C01-EX4	C01-E4, p. 5
BO055	Determining the direction of variation (Skill)	C01-U3, p. 3 ; C01-EX3	C01-E6, p. 6
BO056	Modelling discrete linear or exponential growth (Skill)	C01-U3, p. 3 ; C01-EX3	C01-E6, p. 6
BO057	Intuitive limit: finite, infinite, or no limit (Content)	C01-U5, p. 4 ; C01-EX5	C01-E6, p. 6

Curriculum coverage matrix

ID	Expectation / type	Lesson, example	Exercise
BO058	Conjecturing a limit in simple cases (Skill)	C01-U5, p. 4 ; C01-EX5	C01-AP5, p. 4
BO059	Factorised, expanded and canonical forms (Content)	C02-U1, p. 14 ; C02-EX1	C02-E1, p. 16
BO060	Roots and choice of form (Content)	C02-U1, p. 14 ; C02-EX1	C02-E1, p. 16
BO061	Sum and product of roots (Content)	C02-U3, p. 15 ; C02-EX3	C02-E3, p. 16
BO062	Discriminant, solving and possible factorisation (Content)	C02-U2, p. 14 ; C02-EX2	C02-E3, p. 16
BO063	Studying the sign of a factorised form (Skill)	C02-U4, p. 15 ; C02-EX4	C02-E5, p. 17
BO064	Choosing a form for an equation, inequality, optimisation or variations (Skill)	C02-U4, p. 15 ; C02-EX4	C02-E5, p. 17
BO065	Finding the polynomials vanishing at two distinct real numbers (Skill)	C02-U3, p. 15 ; C02-EX3	C02-E4, p. 17
BO066	Diversifying factorisation strategies (Skill)	C02-U3, p. 15 ; C02-EX3	C02-E4, p. 17
BO067	Solving the quadratic equation by proof by cases (Proof)	C02-U2, p. 14 ; C02-EX2	C02-E6, p. 17
BO068	Rate of change and secant lines (Content)	C04-U1, p. 36 ; C04-EX1	C04-E1, p. 38
BO069	Derivative at a point as the intuitive limit of the rate (Content)	C04-U2, p. 36 ; C04-EX2	C04-E2, p. 38
BO070	Tangent as the limit of secant lines, and gradient (Content)	C04-U3, p. 37 ; C04-EX3	C04-E5, p. 39
BO071	Tangent linear approximation (Content)	C04-U3, p. 37 ; C04-EX3	C04-E5, p. 39
BO072	Calculating a rate and the gradient of a secant (Skill)	C04-U1, p. 36 ; C04-EX1	C04-E1, p. 38
BO073	Interpreting gradient, instantaneous velocity and marginal cost (Skill)	C04-U2, p. 36 ; C04-EX2	C04-E3, p. 38
BO074	Determining a derivative at a point graphically (Skill)	C04-U3, p. 37 ; C04-EX3	C04-E4, p. 39
BO075	Constructing a tangent with a given gradient (Skill)	C04-U3, p. 37 ; C04-EX3	C04-E4, p. 39
BO076	Determining the equation of a tangent (Skill)	C04-U3, p. 37 ; C04-EX3	C04-E4, p. 39
BO077	Calculating an approximation of f(a+h) (Skill)	C04-U3, p. 37 ; C04-EX3	C04-E4, p. 39
BO078	Equation of the tangent (Proof)	C04-U3, p. 37 ; C04-EX3	C04-E2, p. 38
BO079	Square root not differentiable at 0 (Proof)	C04-U4, p. 37 ; C04-EX4	C04-AP4, p. 38
BO080	Differentiable function and derivative function (Content)	C06-U1, p. 59 ; C06-EX1	C06-E1, p. 61
BO081	Derivatives of the square, cube, inverse and square root (Content)	C06-U1, p. 59 ; C06-EX1	C06-E1, p. 61
BO082	Derivative of a positive or negative integer power (Content)	C06-U1, p. 59 ; C06-EX1	C06-E1, p. 61
BO083	Sum, product, reciprocal and quotient of differentiable functions (Content)	C06-U2, p. 59 ; C06-EX2	C06-E1, p. 61
BO084	Absolute value: differentiability at 0 (Content)	C04-U4, p. 37 ; C04-EX4	C04-AP4, p. 38
BO085	Calculating a derivative function using the operations (Skill)	C06-U2, p. 59 ; C06-EX2	C06-E1, p. 61
BO086	Derivative of the square function (Proof)	C06-U1, p. 59 ; C06-EX1	C04-E8, p. 40

ID	Expectation / type	Lesson, example	Exercise
BO087	Derivative of the inverse function (Proof)	C06-U1, p. 59 ; C06-EX1	C04-E8, p. 40
BO088	Derivative of a product (Proof)	C06-U2, p. 59 ; C06-EX2	C06-E1, p. 61
BO089	Even and odd functions, graphical interpretations (Content)	C06-U4, p. 60 ; C06-EX4	C06-AP4, p. 61
BO090	Link between the sign of the derivative and variations on an interval (Content)	C06-U3, p. 60 ; C06-EX3	C06-E6, p. 63
BO091	Characterisation of constant functions (Content)	C06-U3, p. 60 ; C06-EX3	C06-E6, p. 63
BO092	Derivative at a point and tangent at an extremum (Content)	C06-U3, p. 60 ; C06-EX3	C06-E6, p. 63
BO093	Studying variations and extrema (Skill)	C06-U3, p. 60 ; C06-EX3	C06-E2, p. 62
BO094	Studying quadratics using differentiation (Skill)	C06-U3, p. 60 ; C06-EX3	C06-E2, p. 62
BO095	Solving an optimisation problem (Skill)	C06-U5, p. 61 ; C06-EX5	C06-E5, p. 62
BO096	Establishing an inequality (Skill)	C06-U4, p. 60 ; C06-EX4	C06-E3, p. 62
BO097	Studying the relative position of two curves (Skill)	C06-U4, p. 60 ; C06-EX4	C06-E3, p. 62
BO098	Definition by f' equal to f and f(0)=1 (Content)	C09-U1, p. 93 ; C09-EX1	C09-AP1, p. 93
BO099	Existence and uniqueness (accepted without proof) (Content)	C09-U1, p. 93 ; C09-EX1	C09-AP1, p. 93
BO100	The number e and power notation (Content)	C09-U1, p. 93 ; C09-EX1	C09-AP1, p. 93
BO101	Addition of arguments and product (Content)	C09-U2, p. 93 ; C09-EX2	C09-E1, p. 96
BO102	Inverse, quotient and powers (Content)	C09-U2, p. 93 ; C09-EX2	C09-E1, p. 96
BO103	Sign, direction of variation and curve (Content)	C09-U2, p. 93 ; C09-EX2	C09-E1, p. 96
BO104	Link with geometric sequences (Content)	C09-U4, p. 94 ; C09-EX4	C09-E8, p. 97
BO105	Transforming an exponential expression (Skill)	C09-U2, p. 93 ; C09-EX2	C09-E1, p. 96
BO106	Differentiating exp(at) (Skill)	C09-U3, p. 94 ; C09-EX3	C09-E3, p. 96
BO107	Representing exp(kt) and exp(-kt) for positive k (Skill)	C09-U4, p. 94 ; C09-EX4	C09-E4, p. 96
BO108	Modelling exponential growth or decay (Skill)	C09-U4, p. 94 ; C09-EX4	C09-E4, p. 96
BO109	Unit circle, arc and radian (Content)	C10-U1, p. 105 ; C10-EX1	C10-E1, p. 108
BO110	Wrapping the real line and image of a real number (Content)	C10-U2, p. 105 ; C10-EX2	C10-E2, p. 108
BO111	Sine, cosine, right-angled triangle and special values (Content)	C10-U3, p. 106 ; C10-EX3	C10-E3, p. 108
BO112	Placing a point on the circle (Skill)	C10-U2, p. 105 ; C10-EX2	C10-E2, p. 108
BO113	Reading associated angles and their sines and cosines (Skill)	C10-U4, p. 106 ; C10-EX4	C10-E3, p. 108
BO114	Cosine and sine of $\frac{\pi}{4}$ (Proof)	C10-U3, p. 106 ; C10-EX3	C10-RECH, p. 105
BO115	Cosine and sine of $\frac{\pi}{3}$ (Proof)	C10-U3, p. 106 ; C10-EX3	C10-RECH, p. 105

Curriculum coverage matrix

ID	Expectation / type	Lesson, example	Exercise
BO116	Dot product by projection and cosine (Content)	C03-U1, p. 25 ; C03-EX1	C03-E4, p. 28
BO117	Characterisation of orthogonality (Content)	C03-U1, p. 25 ; C03-EX1	C03-E4, p. 28
BO118	Expression in an orthonormal basis and norm (Content)	C03-U2, p. 25 ; C03-EX2	C03-E1, p. 27
BO119	Coordinates via dot products with the basis vectors (Content)	C03-U2, p. 25 ; C03-EX2	C03-E1, p. 27
BO120	Bilinearity and symmetry (Content)	C03-U3, p. 26 ; C03-EX3	C03-E3, p. 28
BO121	Squares of the norms of a sum and a difference (Content)	C03-U3, p. 26 ; C03-EX3	C03-E3, p. 28
BO122	Law of cosines (Al-Kashi formula) (Content)	C03-U4, p. 26 ; C03-EX4	C03-E8, p. 29
BO123	Transformation of MA dot MB (Content)	C03-U5, p. 27 ; C03-EX5	C03-E5, p. 28
BO124	Calculating an angle and a length (Skill)	C03-U4, p. 26 ; C03-EX4	C03-E3, p. 28
BO125	Choosing projection, coordinates, norms and angle (Skill)	C03-U4, p. 26 ; C03-EX4	C03-E3, p. 28
BO126	Solving a geometric problem (Skill)	C03-U4, p. 26 ; C03-EX4	C03-E3, p. 28
BO127	Proving orthogonality (Skill)	C03-U2, p. 25 ; C03-EX2	C03-E2, p. 28
BO128	Law of cosines (Al-Kashi formula) via the dot product (Proof)	C03-U4, p. 26 ; C03-EX4	C03-E8, p. 29
BO129	Locus of MA dot MB equal to zero (Proof)	C03-U5, p. 27 ; C03-EX5	C03-RECH, p. 25
BO130	Normal vector and Cartesian equation (Content)	C07-U1, p. 71 ; C07-EX1	C07-E1, p. 73
BO131	Orthogonal projection onto a line (Content)	C07-U2, p. 71 ; C07-EX2	C07-E2, p. 73
BO132	Equation of a circle (Content)	C07-U3, p. 72 ; C07-EX3	C07-E3, p. 73
BO133	Line through a point with a given normal (Skill)	C07-U1, p. 71 ; C07-EX1	C07-E1, p. 73
BO134	Coordinates of the orthogonal projection (Skill)	C07-U2, p. 71 ; C07-EX2	C07-E2, p. 73
BO135	Equation from the centre and the radius (Skill)	C07-U3, p. 72 ; C07-EX3	C07-E3, p. 73
BO136	Recognising a circle, its centre and its radius (Skill)	C07-U3, p. 72 ; C07-EX3	C07-E3, p. 73
BO137	Using a coordinate system for a configuration (Skill)	C07-U4, p. 72 ; C07-EX4	C07-E4, p. 74
BO138	Independence of two events (Content)	C05-U3, p. 48 ; C05-EX3	C05-E3, p. 50
BO139	Partition of the sample space and total probability (Content)	C05-U2, p. 47 ; C05-EX2	C05-E2, p. 50
BO140	Sequence of two independent trials (Content)	C05-U4, p. 48 ; C05-EX4	C05-E5, p. 50
BO141	Repetition of identical independent Bernoulli trials for n at most 4 (Content)	C05-U4, p. 48 ; C05-EX4	C05-E5, p. 50
BO142	Calculating using total probability (Skill)	C05-U2, p. 47 ; C05-EX2	C05-E2, p. 50
BO143	Using or justifying independence (Skill)	C05-U3, p. 48 ; C05-EX3	C05-E3, p. 50
BO144	Tree or table for two independent trials (Skill)	C05-U4, p. 48 ; C05-EX4	C05-E5, p. 50

ID	Expectation / type	Lesson, example	Exercise
BO145	Tree and calculation for n at most 4 Bernoulli trials (Skill)	C05-U4, p. 48 ; C05-EX4	C05-E5, p. 50
BO146	Random variable as a function on a finite sample space (Content)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO147	Probability distribution (Content)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO148	Expected value (Content)	C08-U2, p. 81 ; C08-EX2	C08-E5, p. 85
BO149	Linearity of expected value (Content)	C08-U2, p. 81 ; C08-EX2	C08-E5, p. 85
BO150	Variance and standard deviation (Content)	C08-U3, p. 82 ; C08-EX3	C08-E2, p. 84
BO151	König–Huygens formula (Content)	C08-U3, p. 82 ; C08-EX3	C08-E2, p. 84
BO152	Interpreting the events X=a and X less than or equal to a (Skill)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO153	Moving between natural language and symbols (Skill)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO154	Modelling with a random variable (Skill)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO155	Determining a distribution (Skill)	C08-U1, p. 81 ; C08-EX1	C08-E1, p. 83
BO156	Calculating expected value, variance and standard deviation (Skill)	C08-U3, p. 82 ; C08-EX3	C08-E2, p. 84
BO157	Using the expected value for a fair game (Skill)	C08-U2, p. 81 ; C08-EX2	C08-E3, p. 84
BO158	Simulating a random variable (Skill)	C08-U5, p. 83 ; C08-EX5	TP, p. 118
BO159	Writing the mean of a sample (Skill)	C08-U5, p. 83 ; C08-EX5	TP, p. 118
BO160	Comparing simulated mean and expected value (Skill)	C08-U5, p. 83 ; C08-EX5	TP, p. 118
BO161	Simulating N samples of size n and the frequency within 2 sigma over root n (Skill)	C08-U5, p. 83 ; C08-EX5	TP, p. 118
BO162	Terms, sums and thresholds (Algorithmic example)	C01-U5, p. 4 ; C01-EX5	TP, p. 118
BO163	Factorial, Fibonacci and Collatz (Algorithmic example)	PYTHON, p. 117 ; PYTHON	TP, p. 118
BO164	List of the gradients of secant lines (Algorithmic example)	C04-U1, p. 36 ; C04-EX1	TP, p. 118
BO165	Newton's method in a favourable case (Algorithmic example)	C06-U5, p. 61 ; C06-EX5	TP, p. 118
BO166	Euler's method and approximation of e (Algorithmic example)	C09-U5, p. 95 ; C09-EX5	TP, p. 118
BO167	Approximation of pi by Archimedes (Algorithmic example)	C10-U5, p. 107 ; C10-EX5	TP, p. 118
BO168	Monte Carlo method: parabola and disc (Algorithmic example)	C05-U5, p. 49 ; C05-EX5	TP, p. 118
BO169	Expected value, variance and standard deviation in Python (Algorithmic example)	C08-U3, p. 82 ; C08-EX3	TP, p. 118
BO170	Letter frequencies (Algorithmic example)	C08-E7, p. 85 ; C08-E7	TP, p. 118